

Xiaotong Luo

☎ (+852) 9824-7803 | 🏠 xiaotongtt | ✉ xiaotongtt123@gmail.com | in xiaotongluo | 🎓 Google Scholar

ABOUT ME

Postdoctoral researcher specializing in embodied AI, generative models, and efficient deep learning. Currently developing VLA-based robotic manipulation policies at an embodied AI startup, with end-to-end experience spanning UMI demonstration collection, policy training, and real-robot deployment. Published 20+ papers in leading venues including TPAMI, CVPR, ECCV, NeurIPS, AAAI, and ACM MM, with a strong track record in multimodal learning, diffusion models, and efficient inference.

EDUCATION

Xiamen University	Xiamen, China
Ph.D. in Computer Science and Technology	Sep. 2020 – Jun. 2024
Xiamen University	Xiamen, China
M.Eng. in Computer Science and Technology	Sep. 2017 – Jun. 2020
Xiamen University	Xiamen, China
B.Eng. in Electronic Science and Engineering	Sep. 2013 – Jun. 2017

TECHNICAL SKILLS

Machine Learning & AI: Embodied AI, Robot Learning, Vision-Language-Action Models, Imitation Learning, Generative AI, Computer Vision

Embodied Data & Deployment: UMI Demonstration Collection, VLA Policy Training, Real-Robot Deployment and Evaluation, Data and Temporal Alignment

Model Optimization & Deployment: Inference Acceleration, Model Compression (Distillation, Pruning, Sparsity), Latency Optimization, Efficient Training, LoRA Fine-tuning

Programming & Frameworks: Python, C/C++, PyTorch, TensorFlow, Hugging Face

WORK EXPERIENCE

[Embodied AI][VLA] UMI-to-Real-Robot Policy Generalization

Embodied AI Algorithm Intern, Embodied AI Startup Jul. 2026 – Present

- Owned an end-to-end UMI-based robot learning pipeline spanning demonstration collection and processing, VLA policy training, real-robot deployment, and evaluation.
- Trained and deployed VLA policies based on $\pi_{0.5}$ (pi0.5), aligning data dimensions and temporal resolutions between UMI demonstrations and the robot execution interface.
- Improved the real-robot task success rate from **50%** to **75%** through systematic data and temporal alignment, a gain of **25 percentage points**.

[VLM][3DGS] Scene Synthesis with 3DGS from Few Images

Postdoctoral Researcher, Hong Kong Polytechnic University Hong Kong, China Sep. 2024 – Present

- Developed a few-shot underwater scene reconstruction system using 3D Gaussian Splatting and diffusion priors, improving visual fidelity and geometric consistency under sparse-view constraints.
- Designed depth-guided supervision and multi-view consistency optimization to reduce ambiguity in sparse data settings.
- Integrated diffusion-based perceptual enhancement to recover fine-grained textures, improving realism in challenging underwater scenarios.

[Imaging Systems][ISP] Video Super-Resolution and Raw Image Denoising

Research Engineer, OmniVision Ningbo, China Oct. 2025 – Present

- Designed and optimized an RGB-IR imaging pipeline covering demosaicing, denoising, and low-light enhancement for real-world sensor systems.
- Conducted chip-level evaluation and sensor pattern optimization, identifying efficient configurations under hardware constraints.
- Developed and validated ISP algorithms under hardware constraints, supporting deployment on real devices.
- Led an intern team in the NTIRE challenge and achieved competitive benchmark performance.

[AIGC][Diffusion] Diffusion Inference with Reinforcement Learning

Shenzhen, China

AIGC Algorithm Intern, Noah's Ark Laboratory

Jun. 2022 – Jun. 2023

- Developed a reinforcement learning-based (PPO) diffusion acceleration method, significantly reducing inference latency (to 0.69s) while maintaining high perceptual quality.
- Optimized compute efficiency (2,800+ GFLOPs) for large-scale diffusion models, enabling deployment under constrained compute environments.
- Applied the accelerated model to a real-world invoice recognition pipeline, improving robustness on degraded inputs (wrinkles, noise, artifacts).
- Transferred the solution to production teams and filed a company patent; published at AAAI 2024.

SELECTED PROJECTS

[AIGC][Diffusion] Efficient Diffusion-based Image Restoration

- Built an efficient diffusion framework for image restoration using LoRA fine-tuning and degradation-aware feature modulation, reducing inference cost while maintaining high-quality outputs.
- Designed a one-step sampling strategy to accelerate inference, improving efficiency for real-world deployment scenarios.
- Developed a unified architecture capable of handling multiple degradation types, improving generalization across datasets.

[VLM][Diffusion] Multimodal Diffusion for Underwater Image Enhancement

- Developed a multimodal image enhancement system combining vision-language models and diffusion generators to improve color fidelity and detail restoration.
- Designed semantic-guided color correction using LLaVA-based feature extraction, improving robustness under complex degradation conditions.
- Applied the system to challenging underwater scenarios, improving visual realism and consistency.

[Efficient AI] Model Compression and Acceleration for Image/Video Processing Sep. 2022 – Feb. 2024

- Developed adaptive token sparsification and dynamic pruning methods, reducing computation by up to 90
- Designed dynamic routing and early-exit mechanisms to skip redundant computation, achieving up to 50
- Built scalable models that adapt to different memory and latency constraints, enabling deployment across diverse hardware settings.

SELECTED PUBLICATIONS

[Journal] “Lattice Network for Lightweight Image Restoration.” X. Luo, Y. Xie, Y. Zhang, Y. Qu, C. Li, Y. Fu. *TPAMI* 2023.

[Journal] “Farewell to CycleGAN: Single GAN with Decoupled Constraint for Unpaired Image Dehazing.” X. Luo, W. Yang, Y. Xie, Y. Qu. *Neurocomputing* 2025.

[Journal] “EdgeFormer: Edge-aware Efficient Transformer for Image Super-resolution.” X. Luo, Z. Ai, Q. Liang, Y. Xie, Y. Qu. *TIM* 2024.

[Journal] “Dehazformer: Nonhomogeneous Image Dehazing with Collaborative Global-Local Network.” X. Luo, W. Yang, Y. Qu, Y. Xie. *Acta Automatica Sinica* 2024.

[Conference] “Diffusion Once and Done: Degradation-Aware LoRA for All-in-One Image Restoration.” N. Tang*, X. Luo*, Z. Cheng, L. Zhou, D. Zhang, Y. Qu. AAAI 2026. (Equal contribution)

[Conference] “AdaFormer: Efficient Transformer with Adaptive Token Sparsification for Image Super-Resolution.” X. Luo, Z. Ai, Q. Liang, Y. Qu, Y. Xie. AAAI 2024. (Oral)

[Conference] “SkipDiff: Adaptive Skip Diffusion Model for High-fidelity Perceptual Image Super-resolution.” X. Luo, Y. Xie, Y. Qu, Y. Fu. AAAI 2024.

[Conference] “SkipVSR: Adaptive Patch Routing for Video Super-Resolution with Inter-Frame Mask.” Z. Ai*, X. Luo*, Y. Qu, Y. Xie. ACM MM 2024. (Equal contribution)

[Conference] “Learning Re-sampling Methods with Parameter Attribution for Image Super-resolution.” X. Luo, Y. Xie, Y. Qu. *NeurIPS* 2023.

[Conference] “Memory-friendly Scalable Super-resolution via Rewinding Lottery Ticket Hypothesis.” J. Lin*, X. Luo*, M. Hong, Y. Qu, Y. Xie, Z. Wu. CVPR 2023. (Equal contribution)

[Conference] “Adjustable Memory-Efficient Image Super-Resolution via Individual Kernel Sparsity.” X. Luo, M. Dai, Y. Zhang, Y. Xie, D. Liu, Y. Qu, Y. Fu, J. Zhang. *ACM MM* 2022.

[Conference] “Boosting Lightweight Single Image Super-resolution via Joint-distillation.” X. Luo, Q. Liang, D. Liu, Y. Qu. *ACM MM* 2021.

[Conference] “LatticeNet: Towards Lightweight Image Super-resolution with Lattice Block.” X. Luo, Y. Xie, Y. Zhang, Y. Qu, C. Li, Y. Fu. *ECCV* 2020.

[Conference] “Deep Wavelet Network with Domain Adaptation for Single Image Demoireing.” X. Luo, J. Zhang, M. Hong, Y. Qu, Y. Xie, C. Li. *CVPRW* 2020. (Winner Award of NTIRE Demoireing)

[Patent] “Efficient Image Super-resolution Based on Adjustable Kernel Sparsity.” Y. Qu, X. Luo, M. Dai, Y. Xie. *Publication Number: CN114820326B*.

[Patent] “Unbalanced Image Super-resolution Based on Bi-sampling Texture-aware Distillation Learning.” Y. Qu, J. Lin, X. Luo, Z. Ai. *Publication Number: CN114881861B*.

HONORS & AWARDS

Honor:

- 2nd Prize of Natural Science Award of Fujian Province (2022)
- Merit Student, Outstanding Student, Xiamen University (2019, 2023, 2024)
- Outstanding Doctor’s and Master’s Thesis of Fujian Province (2021)
- Outstanding Graduates, Xiamen University (2020, 2024)
- Winner Award of CVPR-NTIRE Challenge “Image Demoireing” (2020)

Scholarship:

- National Scholarship for Postgraduate Students (2019, 2023)
- The First Prize Scholarship of Xiamen University (2017–2024)

PROFESSIONAL SERVICES

Reviewer:

- Conference Reviewer: CVPR, ICCV, ACM MM, ECCV, ICLR, ICML, AAAI, NeurIPS, ACCV, ICME
- Journal Reviewer: TPAMI, TIP, TCSVT, Pattern Recognition, TNNLS, TIM, The Visual Computer

Lead Teaching Assistant for Computer Vision:

Sep. 2021 – Jan. 2022

Xiamen University

- Helped develop and teach Computer Vision (AI Systems Senior Design)
- Independently hosted lab sessions and mentored two teams
- Designed three lab assignments involving classical ML algorithms, face detection, and recognition